

Section 1

A worksheet is provided at the end of the test to do your work. Completely filling that worksheet will help you solve the problems correctly and will get you partial credit if you make an error part way through that affects later answers. In the distinction between sample and population values, only concern yourself with using the correct formula called for in the question!

For the questions in this section, use the following data.

Eight teams are competing for the post season lineup in StatsBall. You are working for a sports book and your boss would like some preliminary statistics before they try to set the line. You have been assigned to process the data on percent of games won.

Team	Winning Percentage
Tigers	66
Mustangs	40
Owls	43
Aggies	62
Cougars	66
Longhorns	31
Wildcats	35
Razorbacks	56

1. How many observations are there? 8
2. What is the mean winning percentage? 49.9
3. What is sample variance winning percentage? 203.8
4. What is the population variance winning percentage? 178.4
5. What is the sample standard deviation? 14.3
6. What is the population standard deviation? 13.8
7. What is the median? 49.5
8. What is the mode? 66

Section 2: Multiple Choice

Choose the appropriate answer.

9. Given the following values, what is the mean? (5,10,15,15,20,25) D

a 20

- b 18
 - c 19
 - d 15
10. Given the values in question 9, what is the mode? B
- a 10
 - b 15
 - c 20
 - d 25
11. The principle that allows us to apply the normal distribution to the *central tendency* of large sample sizes regardless of the underlying distribution is the: C
- a 65-95-99.7 rule
 - b The Law of Large Numbers
 - c The Central Limit Theorem
 - d Krusty Krab is Unfair
12. the principle that allows us to apply the expected mean of the sample to the population for *large* enough sample sizes is: B
- a The 65-99-99.7 rule
 - b The Law of Large Numbers
 - c The Central Limit Theorem
 - d The Complement Rule
13. If two sets, A and B, are **not** mutually exclusive, what do we need to subtract from $P(A) + P(B)$ to get $P(A \cup B)$: C
- a $P(A) \cdot P(B)$
 - b $P(C)$
 - c $P(A \cap B)$
 - d $1/2$
14. If two sets, A and B, **are mutually exclusive** and we know the probability of either occurring, $P(A \cup B)$, is 70% and the probability of B occurring, $P(B)$, is 30%, what is the probability of A? D

- a 70%
- b 30%
- c 50%
- d 40%

15. The degree to which two variables move in coordination with each other is: C

- a variance
- b central tendency
- c correlation
- d standard deviation

16. If numbers are entered as a character variable, what function would I need to use to be able to perform math operations on the variable? c - `as.numeric()`

17. If we assign numbers to represent hair color so that 1 = red, 2 = brown, 3 = blonde, 4 = black, 5 = gray, 6 = other, this is which type of categorical variable: B

- a Ordinal
- b Nominal
- c Something else
- d Discrete

18. If we assign numbers to represent ranking survey choices in order from worst to best, this is which type of categorical variable: A

- a Ordinal
- b Nominal
- c Something else
- d Continuous

3: Fill in the blank/short answer

Provide the appropriate answer. One or two words for fill in the blank and just a few words for short answer unless otherwise noted.

19. In R: Named vectors and dataframes are examples of OBJECTS, items that have been named, assigned values or meanings, and stored in the Environment for R to use.
20. In R: If I want to find the mean of variable X from dataframe *my_data* I would type `mean(my_data __ X)`. (The blank is a single letter, number or symbol that goes between the dataframe name and the variable name.) \$
21. The mean, median, and mode are measures of CENTRAL tendency.
22. The variance and STANDARD DEVIATION both measure difference from the mean.
23. Of the variance and the standard deviation, which has the same scale or units as the original data? STANDARD DEVIATION
24. The advantage of the MEDIAN is that it is less influenced by a single extreme observation.
25. The MODE is the only measure of central tendency that can be used for categorical variables.
26. The 68-95-99.7 rule tells us that 68% of outcomes fall within 1 STANDARD DEVIATION of the mean in a normal distribution, 95% fall within two, and 99.7% fall within three.
27. Two events are mutually EXCLUSIVE if they share no common elements, if they have zero probability of occurring at the same time.
28. The probability that event A occurs given that event B occurs is a conditional probability. To compute this sort of probability you would apply BAYES Rule.
29. The complement of an event is when the event DOES NOT HAPPEN . (Best answer is two-three words.)
30. The UNION of two sets, A and B, is a set containing all elements that are in A or in B (possibly both).
31. $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$
32. $a = \{HHH\}$
33. for "at least" two tails: $b = \{HTT, THT, TTH, TTT\}$; for "exactly" two tails $b = \{HTT, THT, TTH\}$
34. Independent

4: Calculations

Partial work is possible for properly setting up the problem. Most of these are a single math operation (addition, division, etc.) and can be done on a

calculator if needed. Simplify fractions, but you can answer with fractions or decimals. You do not need to compute square roots. For example, $\frac{1}{2}$ and $\sqrt{10}$ are acceptable answers. $\frac{5}{10}$ and $\sqrt{10+10+10}$ are not.

35. If the sum of the squared deviations from the mean is 2,500,000 and the number of observations is 100,000, what is the population variance? $2,500,000/100,000 = 25$

36. If the sum of the squared deviations from the mean is 2,500,000 and the number of observations is 100,000, what is the population standard deviation? $\sqrt{25} = 5$

37. If the mean is 50 and the standard deviation is 10 for a normally distributed population, what proportion (percentage) of the outcomes should fall between 40 and 60? Between 30 and 70? 68% and 95%. 40 to 60 is 1 within 1 standard deviation of 50. 30 to 70 is within two standard deviations of 50. Apply the 68-95-99 rule.

38. If the sum of the observations is 1000 and the number of observations is 100, what is the mean? 10

39. If the population variance is 500 and the number of observations is 50, what is the sample variance? $500 * (50/49) = 510.2$

40. If the population variance is 500 and the number of observations is 50, what is the population standard deviation? $\sqrt{500} = 22.36$

41. If the mean is 100 and the standard deviation is 10, 99.7% of observations should be between what two values? $100 \pm 3 * 10$ between 70 and 130

42. If the sum of the squared deviations from the mean is 500 and the number of observations is 50, what is the population standard deviation? $\sqrt{500/50} = \sqrt{10}$

43. Question 31 had a sample space with 8 outcomes. Question 32 had a favorable outcomes equal to 1. $\frac{\text{favorable outcomes}}{\text{possible outcomes}} = 1/8$

Team (Observation)	Wins percentage (Value)	Deviations from the Mean	Squared Deviations from the Mean
	x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
Item	Symbol	<i>Value:</i>	
Sum	Σx_i		
Observations	n		
Observations - 1	$n - 1$		
Mean	$\bar{x} = \frac{\Sigma x_i}{n}$		
Sum Squared Deviations	$\Sigma(x_i - \bar{x})^2$		
Sample Variance	$\frac{\Sigma(x_i - \bar{x})^2}{n-1}$		
Sample Standard Deviation	$\sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n-1}}$		
Population Variance	$\frac{\Sigma(x_i - \bar{x})^2}{n}$		
Population Standard Deviation	$\sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n}}$		
Median	Middle or $\frac{lower\ middle + higher\ middle}{2}$		
Mode	Just count		

Formulas

Mean	$\bar{x} = \frac{\sum x_i}{n}$	
Sum Squared Deviations	$\sum (x_i - \bar{x})^2$	
Sample Variance	$\frac{\sum (x_i - \bar{x})^2}{n-1}$	
Sample Standard Deviation	$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$	
Population Variance	$\frac{\sum (x_i - \bar{x})^2}{n}$	
Population Standard Deviation	$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$	
Median	Middle or $\frac{\text{lower middle} + \text{higher middle}}{2}$	
Mode	Just count	
Probability Formulas:		
P(A)	$P(A) = \frac{\text{favorable outcomes}}{\text{possible outcomes}}$	
Mutually exclusive		
$P(A \cap B)$	0	
$P(A \cup B)$	$P(A \cup B) = P(A) + P(B)$	
Non-Mutually exclusive		
$P(A \cup B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	